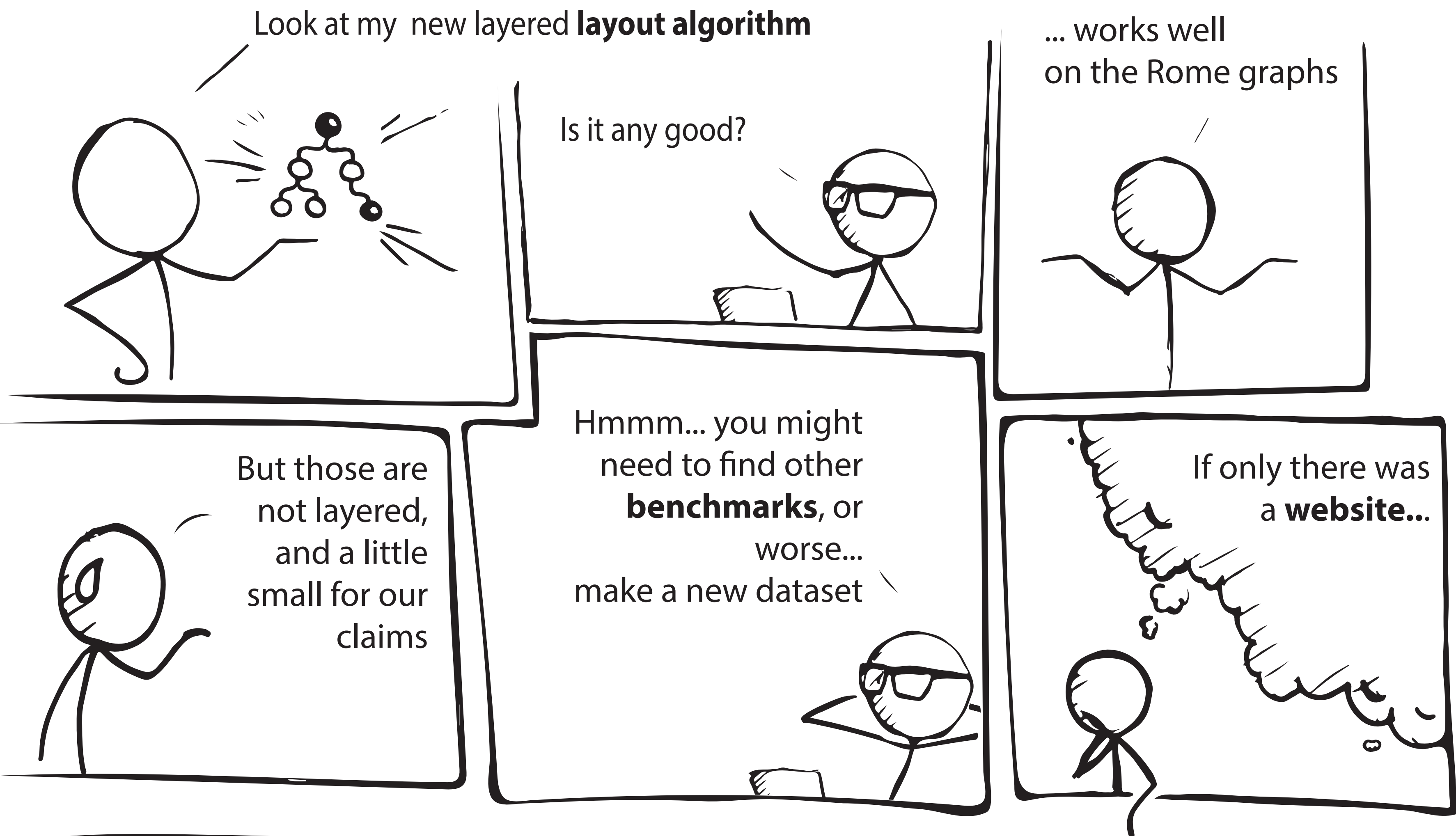


Graph Drawing Benchmark Datasets: A History, Classification, and Collection

Sara Di Bartolomeo, Eduardo Puerta, Connor Wilson, Tarik Crnovrsanin, and Cody Dunne



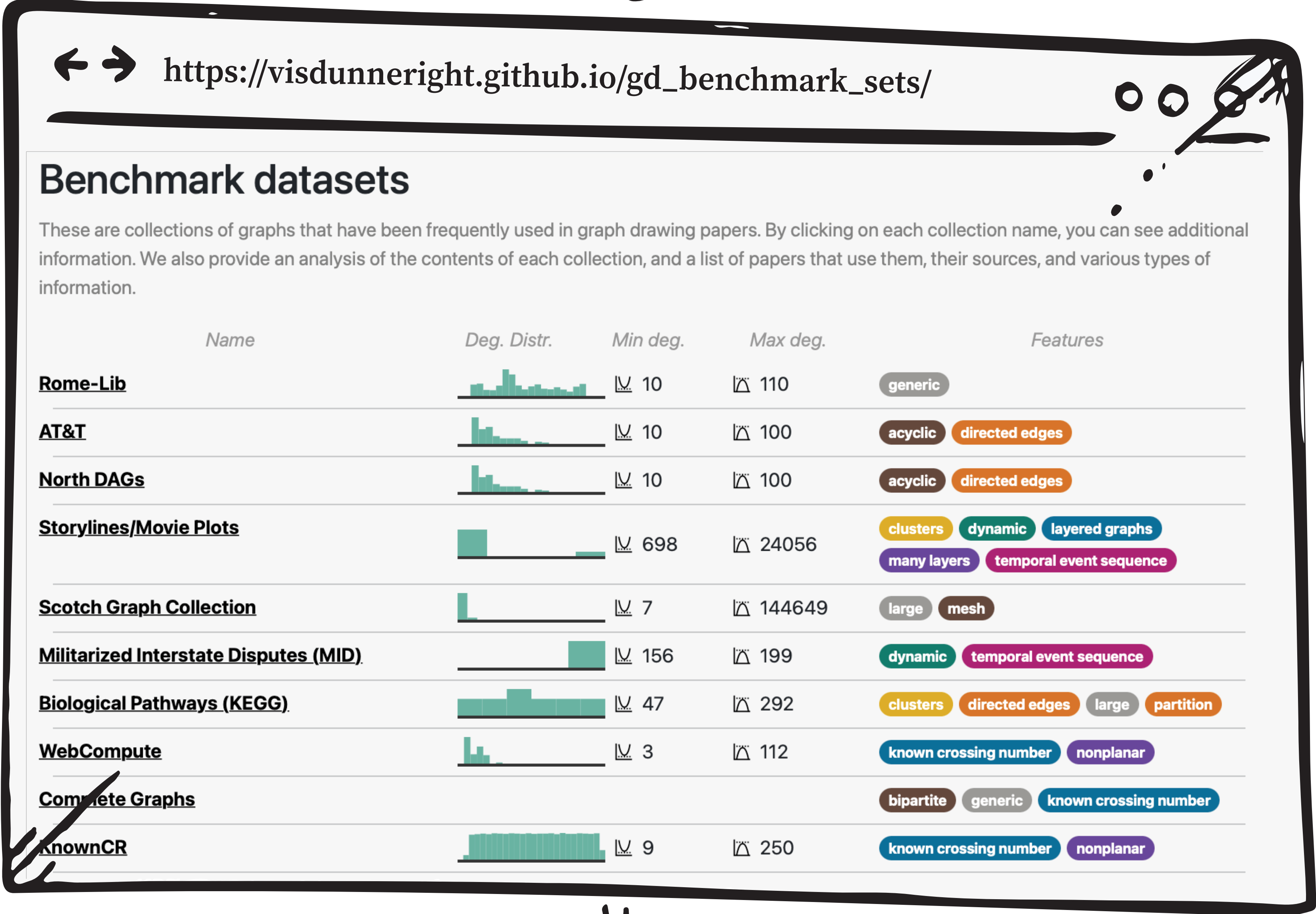
Have you ever struggled finding a graph drawing dataset?

We made a website just for you!

Benchmarking provides a standardized way to compare and assess the performance of different systems, algorithms, or software implementations. In **Graph Drawing**, we use them to study the performance and results of graph layout algorithms. These benchmarks depend on the properties of the graphs to be laid out, which can vary in size, density, connectivity, and structure.

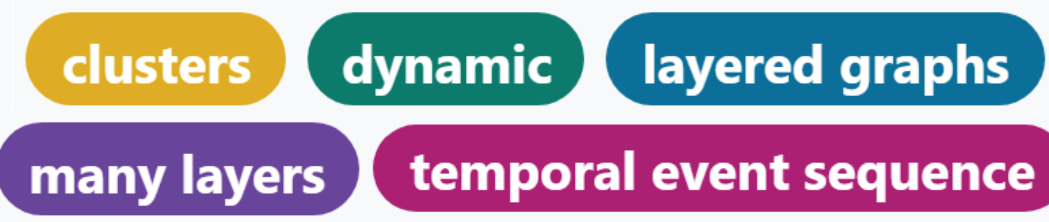
From personal experience, finding benchmark sets that deal with specific **graph features** is complicated. That is why we compiled and classified some of the most relevant benchmarks for graph drawing on our **website**! Next time you are designing a layout algorithm for a specific type of graph you have a database to test your algorithm on!

While there are many existing network repositories, like the SuiteSparse Matrix Collection [1], we focus specifically on benchmarks used in graph drawing research. However, we also link to many of these repositories when relevant.

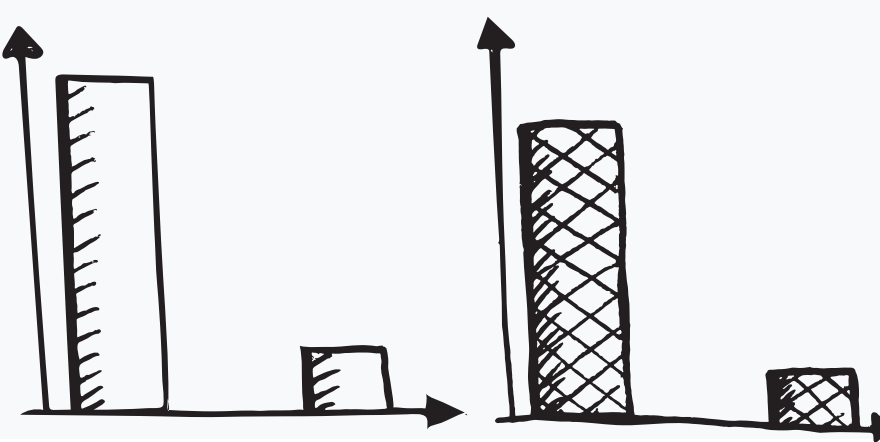


A website...

That characterized benchmarks in the **graph drawing** literature by **graph features**



Showed **relevant statistics** like number of nodes or degree distribution for the graphs



Archived datasets in multiple **standard formats** in OSF for long term preservation and to support **reproducible research**



Has **example figures** of how others use the graphs

From "StoryFlow: Tracking the Evolution of Stories" [2] :

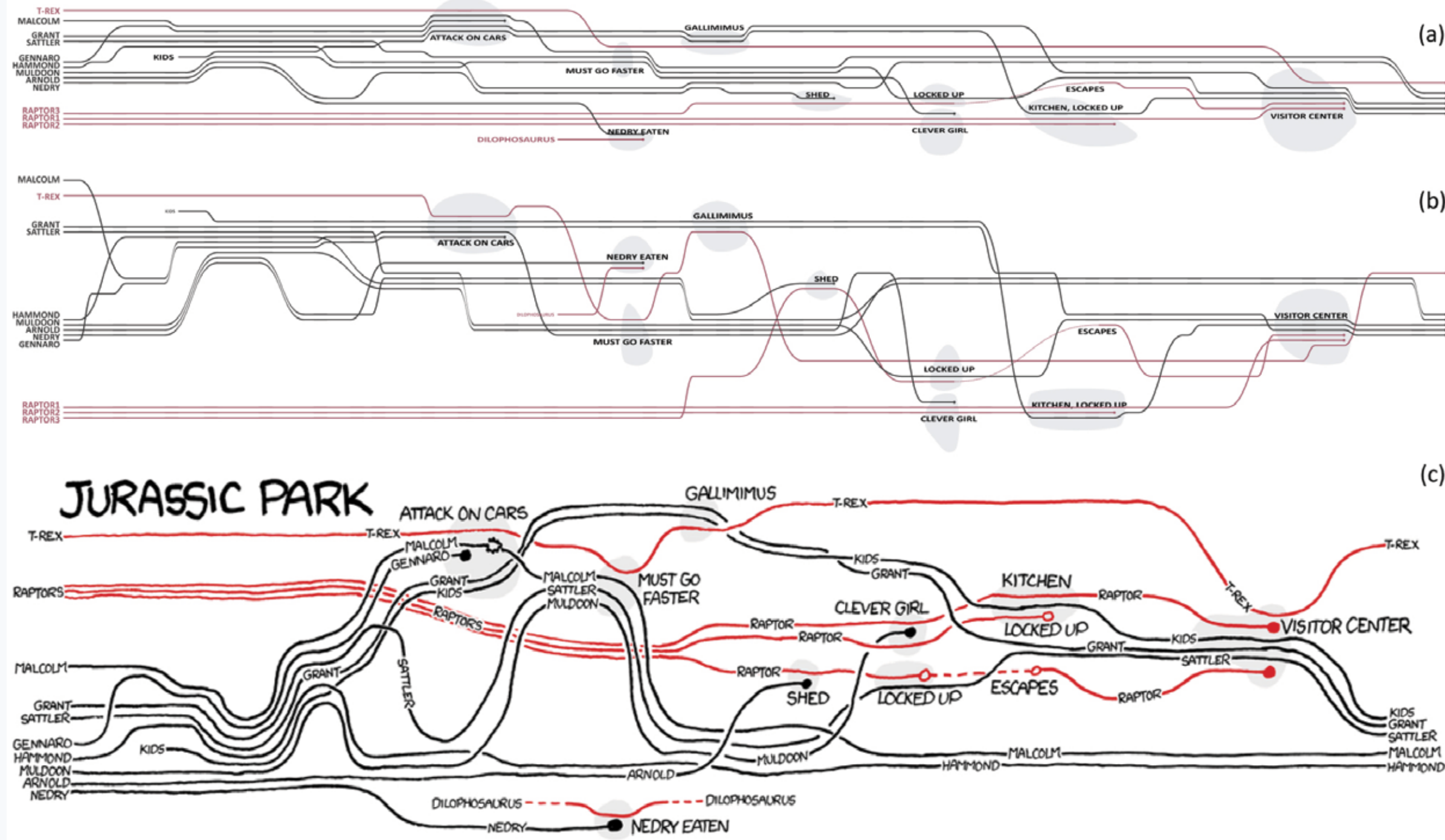


Fig.11. Comparison of the movie Jurassic Park: (a) layout by StoryFlow ; (b) layout by the TM method; (c) hand-drawn illustration from XKCD.

Shows details for each of the benchmarks, like its **origin**, usages, and interesting notes

Name: AT&T

Node Distr. [Histogram]

Min nodes: 10

Max nodes: 100

Features: acyclic, directed edges

Originally found at: <http://www.graphdrawing.org/data.html>

Download formatted data: [GEXF](#) [GML](#) [GRAPHML](#) [JSON](#)

Note: Same as North DAG collection.

Size: 1277 graphs, 10 to 100 nodes, 9 to 241 edges

Origin Notes: Originally collected by Stephen North at AT&T Bell Labs. The original link from 1995 is broken: <ftp://ftp.research.att.com/dist/drawdag>. DiBattista et al. proceeded to remove isomorphic graphs, connecting disconnected graphs and remove cycles with heuristics.

Origin paper: [\[link\]](#) [\[citation\]](#) Drawing Directed Acyclic Graphs: An Experimental Study

Usage examples:

- [\[link\]](#) Drawing Large Graphs with a Potential-Field-Based Multilevel Algorithm
- [\[link\]](#) An Experimental Comparison of Fast Algorithms for Drawing General Large Graphs
- [\[link\]](#) Drawing Large Graphs with a Potential-Field-Based Multilevel Algorithm
- [\[link\]](#) Large-Graph Layout with the Fast Multipole Multilevel Method
- [\[link\]](#) Compact Layered Drawings of General Directed Graphs
- [\[link\]](#) A Flow Formulation for Horizontal Coordinate Assignment with Prescribed Width

[\[link\]](#) A Natural Quadratic Approach to the Generalized Graph Layering Problem

[\[link\]](#) Advances in the Planarization Method: Effective Multiple Edge Insertions

[\[link\]](#) Simple and Efficient Bilayer Cross Counting

[\[link\]](#) Drawing Directed Acyclic Graphs: An Experimental Study

[\[link\]](#) Measuring Symmetry in Drawings of Graphs

generic large

generic large

generic large

generic large

directed edges layered graphs n-layers

dag hierarchical layered graphs

n-layers

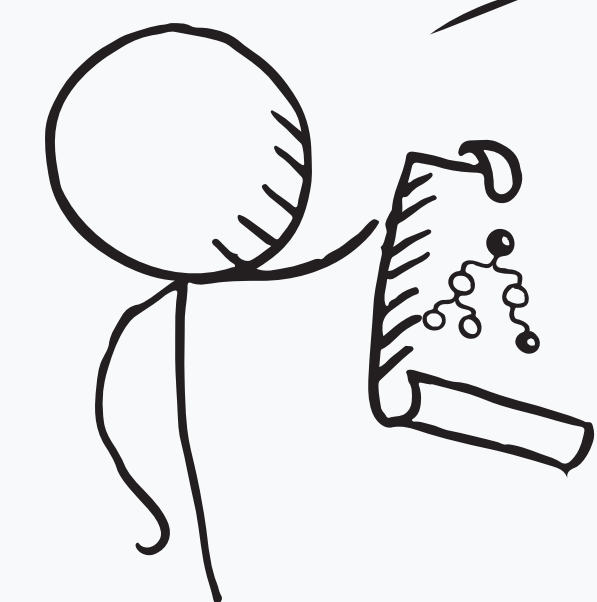
layered graphs n-layers

generic

bipartite layered graphs

directed edges layered graphs n-layers

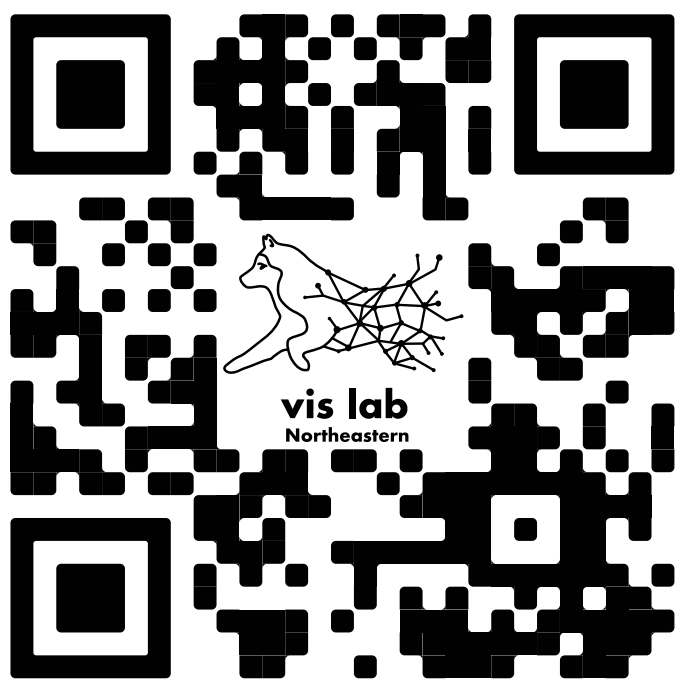
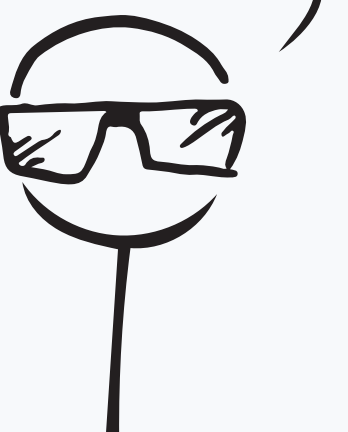
generic



IT EXISTS! This dataset is perfect

Now my paper can't be rejected!

I don't think that's how it works



<https://osf.io/j7ucv/>

https://visdunneright.github.io/gd_benchmark_sets/

References

- [1] Davis, T., & Hu, Y. (2011). The University of Florida Sparse Matrix Collection. ACM Transactions on Mathematical Software 38, 1, doi: <https://doi.org/10.1145/2049662.2049663>
- [2] Liu, S., Wu, Y., Wei, E., Liu, M., & Liu, Y. (2013). StoryFlow: Tracking the Evolution of Stories. IEEE Transactions on Visualization and Computer Graphics, 19(12), 2436–2445. doi:10.1109/TVCG.2013.196